PCSPF

USER & IMPLEMENTATION GUIDE

Port-Corridor SME Performance Framework

Operations Toolkit v1.0 | Microsoft Excel | U.S. market conventions

Financial controls / Dispatch reliability / Equipment costs / Data integrity

Illustrative dashboard • actual workbook rendering



RELEASE CLASSIFICATION

Synthetic computational prototype. The 100 shipments, equipment logs, financial values and scenario comparisons are illustrative only. They are not observations from U.S. firms, verified savings or evidence of independent adoption.

Version 1.0 • November 9, 2026 • Prepared for technical review

Project owner: Ricardo Lopes Wiest | Companion file: PCSPF_v1_0_US_Operations_Toolkit.xlsx

01 Document control and permitted uses

Control item	Released value
Product	PCSPF Operations Toolkit — synthetic demonstration, version 1.0
Workbook	PCSPF_v1_0_US_Operations_Toolkit.xlsx
Publication date	November 9, 2026
Project owner	Ricardo Lopes Wiest
Intended audience	Small-business operations managers, financial controllers, logistics reviewers and independent technical evaluators.
Model classification	Prototype / training / technical inspection; NOT production-validated.
Source specification	PCSPF Professional Plan, especially §§4, 15, 16, 22–23.
Independent review	Not performed as of this release.

Scope

PCSPF is designed to connect load-level operating events with billing records, direct costs, dispatch duration and basic equipment metrics. Version 1.0 demonstrates the mechanics on fabricated records representing crushed stone, manufactured sand, aggregate base and riprap. Its intended application is freight-adjacent small enterprises—not port infrastructure management or regulated transportation engineering.

What this guide provides

A controlled walkthrough of the workbook; column-by-column input requirements; formula and indicator definitions; expected verification results; known design limitations; and a proposed protocol for a future authorized field evaluation.

What it does not certify

EVIDENCE BOUNDARY

This guide and workbook do not establish that PCSPF was piloted in the United States, adopted by third parties, independently validated, endorsed by any federal agency or responsible for realized economic savings.

02 Quick-start procedure

Step 1 — Preserve the release. Keep the original workbook unchanged as the 100-record demonstration and QA reference. Duplicate it before changing source fields; name any working copy with the location, purpose, date and version.

Step 2 — Inspect the control assumptions. Open CONTROL. Confirm B6 identifies synthetic data; B7 is the fixed as-of date (November 9, 2026); B8 is the liters-per-U.S.-gallon constant (3.785411784). Do not modify the unit conversion constant.

Step 3 — Check the verification baseline. Open QA TESTS. On the unchanged model, all 13 checks should report PASS. Refer to Section 08 for their numerical expectations.

Step 4 — Trace one transaction. In TRANSACTIONS, inspect row 7: source inputs occupy A:V, while calculated fields occupy W:AL. Compare ticket ID F7 against scenario invoice references G7 and H7; inspect Y7/Z7 turnaround and AC7/AD7 direct contribution.

Step 5 — Interpret summary views. Open DASHBOARD, COST MODEL, DISPATCH ANALYSIS and RECEIVABLES. These views are linked to the transaction ledger. EQUIPMENT LOG is a separate, deliberately unreconciled scenario.

Step 6 — Review the limitations. Read DATA DICTIONARY and TECHNICAL NOTES, particularly the distinction between mathematical alternatives and actual before/after operational evidence.

Step 7 — Plan a controlled field adaptation. Before inserting real data, obtain data-use permission, reconcile the source schema, revise and test the baseline/alternative logic, verify accounting treatment, and document a separate version. Do not treat the demonstration as production-ready.

FIRST-PASS ACCEPTANCE

100 synthetic shipments • 85 baseline matches • 97 alternative matches • 5,885 baseline minutes • 5,383 alternative minutes • 59.0 / 53.5 minute medians • 13 of 13 deterministic checks passing.

03 Workbook navigation

Worksheet	Purpose	Operational note
START HERE	Entry instructions and use boundaries.	Read before changing data.
DASHBOARD	Core counts, matching rates, totals, median durations and chart.	Scenario outputs only.
CONTROL	Dataset classification, fixed review date and unit conversion.	B7 controls receivable aging.
TRANSACTIONS	Load-level records, invoice matching, costs, duration and data flags.	A:V = source; W:AL = calculated.
EQUIPMENT LOG	Independent asset-level fuel, utilization and fuel-cost example.	Not reconciled to shipments.
COST MODEL	Product-level direct contribution and margins.	Not EBITDA or net income.
DISPATCH ANALYSIS	Scenario duration statistics by commodity.	Not measured productivity.
RECEIVABLES	Paid / open / overdue classification and paid collection time.	Based on CONTROL B7.
QA TESTS	13 deterministic acceptance checks.	Reference sample only.
DATA DICTIONARY	Field ownership, units, formulas and eligibility rules.	Primary technical crosswalk.
TECHNICAL NOTES	Model design, compliance and analytical limitations.	Read with this guide.
CHANGE LOG	Version, owner and validation status.	Update every controlled release.

Calculation path

TRANSACTIONS A:V → TRANSACTIONS W:AL → DASHBOARD / COST MODEL / DISPATCH ANALYSIS / RECEIVABLES. Separately, EQUIPMENT LOG A:K → EQUIPMENT LOG L:R. QA TESTS verifies the prescribed synthetic sample in both pathways.

04 Transaction ledger: source fields

Input zone: TRANSACTIONS!A7:V206. Rows 7–106 contain 100 fabricated shipments; rows 107–206 are reserved. Formula zone: W7:AL206. Do not paste over the formula zone.

Columns	Required input / basis	Practical control
A–D	Shipment ID; service date; commodity; asset ID.	Unique ID; exact operating date; controlled product name.
E–H	Quantity (U.S. short tons); load ticket ID; baseline and alternative invoice-ticket references.	Positive recorded quantity; references must map to the same source ticket when matched.
I–K	Arrival, baseline departure and alternative departure timestamps.	Consistent time zone; chronological events; alternatives are synthetic in v1.0.
L–M	Quoted rate (USD per short ton); billed revenue (USD).	Quotes and actual billed totals are distinct values.
N–P	Direct material, freight and other direct costs (USD).	Enter only directly attributable costs; document allocation rules.
Q–R	Fuel (liters); equipment operating hours.	Comparable asset and period; operating hours > 0.
S–U	Invoice date; due date; paid date (blank if unpaid).	Use reliable invoice/payment records; keep due date ≥ invoice date.
V	Source / exception note.	Retain source reference and document unresolved exceptions.

Important editing restrictions

- Editable demonstration inputs are styled in blue; formula outputs are dark. Blue does not constitute an access-control or data-quality guarantee.
- Column C has a predefined commodity list. Changing product taxonomy may require updating the hardcoded commodity labels in COST MODEL and DISPATCH ANALYSIS.
- If only actual baseline data exist, do not invent alternative references or timestamps. Version 1.0 expects two scenarios; absent alternative fields may trigger CHECK flags or misleading alternative metrics. Adapt and retest the schema before real use.
- The model currently contains up to 200 formula-supported transaction rows (7–206). Expanding capacity requires coordinated formula and aggregation updates.

05 Equipment log, units and as-of controls

Equipment log

Input zone: EQUIPMENT LOG!A7:K46. The release contains 12 synthetic equipment-period records, with spare input capacity through row 46. Derived fields occupy columns L:R. The equipment ledger is a separate teaching model and does not reconcile to the shipment-level fuel entries.

Inputs / outputs	Meaning	Workbook control
A–D	Record ID, period start, asset ID and class.	Trace the machine and observation interval.
E–H	Productive hours, idle hours, fuel in liters and tons handled.	No negative values; comparable period and activity.
I–K	Fuel rate USD/L, other variable cost and source note.	Fuel rate is a source input, not pulled from CONTROL B9.
L–R	Total hours; utilization; fuel L/hr; fuel U.S. gal/hr; fuel USD; fuel USD/ton; QA status.	Calculated: do not overwrite.

Unit conventions

Measure	Convention	Do not confuse with
Weight	1 U.S. short ton = 2,000 pounds.	Metric tonne = 1,000 kg.
Fuel	Input in liters; display also in U.S. gallons.	Imperial gallon.
Conversion	1 U.S. gallon = 3.785411784 liters (CONTROL B8).	Rounding the conversion constant.
Money	U.S. dollars (USD).	A validated price forecast or official benchmark.
Time	Minutes, hours and Excel dates/timestamps.	Cross-timezone timestamps without normalization.

REVIEW-DATE DEPENDENCY

CONTROL B7 is a fixed demonstration date, not an automatic TODAY() calculation. Update it deliberately in a real authorized copy to recalculate open/overdue receivables, and record the review date in CHANGE LOG.

06 Indicator definitions and formulas

Measure	Formula / source	Interpretation / caveat
Ticket-match rate	SUM(TRANSACTIONS W) / shipment count; alternative: SUM(X) / shipment count.	Exact ticket-reference match only; not a complete invoice audit.
Turnaround minutes	ROUND((departure − arrival) × 1,440, 0); Y = baseline, Z = alternative.	Negative/impossible records must be flagged, not averaged away.
Scenario difference	SUM(Y − Z) across synthetic records.	The 502-minute delta is programmed and is not observed time saved.
Expected billing	E × L (quantity × quoted unit rate).	Excludes fees not represented in the unit quote.
Pricing variance	M − AA (billed less expected).	Nonzero is a review signal, not proof of error.
Direct contribution	M − (N + O + P).	Excludes fixed overhead, taxes, interest and depreciation; NOT net profit.
Contribution margin	Direct contribution / billed revenue, when billed revenue > 0.	Compare only with compatible cost allocation.
Fuel intensity	Q / R liters per hour, R > 0; gallons/hour = (Q / R) / CONTROL B8.	Asset class and operating definitions must match.
Equipment utilization	Productive hours / (productive + idle hours).	Requires reliable productive/idle classifications.
Receivable status	PAID if U populated; otherwise OVERDUE if due date < CONTROL B7; else OPEN.	As-of-date specific; blank or invalid dates may distort classification.
Paid collection days	Paid date − invoice date; RECEIVABLES averages paid items.	Not Days Sales Outstanding (DSO).

Two accounting boundaries

- Fuel consumption is a separate metric in TRANSACTIONS and does not automatically become a direct-cost expense in COST MODEL. Add fuel to accounting costs only through an explicit, non-duplicative allocation policy and a revised controlled version.
- EQUIPMENT LOG and TRANSACTIONS use independent fictional fuel datasets. Combining their totals would double-count or misstate equipment costs.

07 Reporting views and decision use

DASHBOARD

Shows 100 records, 85.0% baseline ticket match, 97.0% alternative ticket match and 100 data-integrity PASS rows for the synthetic release. The bar chart compares mean scenario turnaround by commodity. It is illustrative, not evidence of operational improvement.

COST MODEL

Aggregates sample shipments, short tons, billed revenue, direct material/freight/other costs, contribution and contribution per short ton by four products. “Contribution” should not be described as EBITDA, GAAP profit, cash generated or a U.S. industry benchmark.

DISPATCH ANALYSIS

Compares counts, base/alternative total minutes, average turnaround and constructed duration changes for each product. Use sum of record-level differences for synthetic arithmetic checks; do not multiply changes in medians by load counts to claim aggregate savings.

RECEIVABLES

Classifies billed loads as PAID, OPEN or OVERDUE, and reports a paid-invoice collection lag. Open balance depends on the fixed CONTROL B7 as-of date. The workbook does not calculate an audited DSO ratio or produce a certified receivables ledger.

EQUIPMENT LOG

Calculates fuel intensity, utilization and fuel cost from an illustrative asset log. Do not infer realized efficiency gains or machine health from its fictional observations.

RESPONSIBLE INTERPRETATION

Every chart and KPI in version 1.0 is a demonstration of calculation mechanics. A change between “base” and “alternative” does not represent a controlled field intervention, return on investment, causal savings, customer satisfaction or national-scale benefit.

08 Verification and acceptance tests

Run QA TESTS on the unchanged synthetic reference file. Thirteen deterministic checks compare observed calculated values against fixed expectations. The presence of 13 PASS statuses demonstrates selected arithmetic and internal consistency only.

Reference test	Expected	What it verifies
Shipment records	100	Reference cohort count.
Baseline matched tickets	85	Exact-match indicator.
Alternative matched tickets	97	Constructed alternative reference.
Baseline duration sum	5,885 min	Row-level duration arithmetic.
Alternative duration sum	5,383 min	Row-level scenario arithmetic.
Baseline median	59 min	Distribution summary.
Alternative median	53.5 min	Distribution summary.
Constructed difference	502 min	Sum of programmed row-level changes.
Duplicate IDs	0	Shipment ID uniqueness among populated rows.
Transaction CHECK flags	0	Selected source-data validation rules.
Equipment CHECK flags	0	Selected equipment-data validation rules.
Product-summed loads	100	Product rollup count.
Paid/open/overdue loads	100	Receivables classification count.

CRITICAL CAUTION

QA TESTS uses fixed expected results for the original synthetic dataset. Once real entries or calculations are changed, the 13 PASS results are no longer an appropriate acceptance criterion. Preserve the reference workbook and establish separate, documented tests for each field deployment.

09 Controlled field-adaptation protocol

The following is a proposed pilot methodology aligned to the PCSPF Professional Plan. It is not a claim that any such activity has occurred. A real company deployment requires a separate approved scope, suitable legal authorization and reliable source records.

Gate	Minimum evidence	Reject / stop condition
G1 | Permission	Written business/data authorization, scope, responsible owner.	No lawful access or insufficient confidentiality terms.
G2 | Data mapping	Data dictionary matched to ERP, ticket, billing and machine logs; unit test.	Unexplained missing source fields or mismatched identifiers.
G3 | Baseline	At least eight consecutive usable baseline weeks where feasible, per plan §16.1.	Unreliable records or unstable indicator definitions.
G4 | Intervention	Documented change approval, safety review and process owner.	Any unsafe workflow or unapproved operational change.
G5 | Follow-up	At least eight consecutive weeks after implementation where feasible; same definitions.	Data windows not comparable or critical confounders ignored.
G6 | Independent check	Reviewer reconstructs indicators, accepts limitations and examines negative outcomes.	Claims cannot be traced to permissioned source data.

Evidence hierarchy

Operational observations and verified input provenance outrank promotional claims. Strong future records include signed scopes, dated baseline dictionaries, complete exception logs, before/after source extracts (appropriately redacted), reviewer comments and independent use of published instructions. An independent evaluation is more probative when the reviewer tests the method and identifies limitations rather than merely praises the applicant.

Proposed external dissemination

A non-confidential implementation guide, field definitions, blank models, test cases and revision history could be made available to other practitioners. Document genuine independent adaptation and use; downloads and page views alone do not demonstrate field-level adoption.

10 Security, limitations and known constraints

Operational and data security

- Collect the minimum data required; use pseudonymous shipment and asset IDs. Do not include employee, driver or customer identifiers in a public edition.
- Keep source files with the authorized operator, apply access controls outside Excel, document retention periods and prohibit unauthorized redistribution.
- Before using employer records, obtain permission for extraction, redaction, external review and any publication of results.
- Prioritize vehicle safety, driver-hours rules, equipment maintenance, legal obligations and safe working conditions over throughput targets.

Known prototype constraints

Constraint	Practical consequence
Two-scenario schema	Alternative invoice/timestamp fields are built for synthetic testing; not ready for a genuine single-period live ledger without revision.
Static reference QA	Expected values intentionally fail when scenario data changes; live deployments need a new test suite.
Limited row capacity	TRANSACTIONS formulas cover rows 7–206; EQUIPMENT LOG rows 7–46.
Incomplete integrity audit	PASS checks selected rules only; it does not independently verify invoices, authorizations, accounting cutoffs or completeness.
Unlinked cost pools	Fleet log and transaction fuel inputs are not reconciled; fuel is not automatically included in direct contribution.
Fixed review date	Overdue status is anchored to CONTROL B7 until explicitly updated.
No external assurance	No independent expert sign-off, real client pilot, economic savings or federal endorsement exists in this release.

11 Troubleshooting and release checklist

Symptom	Initial action	Control decision
QA TESTS = FAIL	Confirm you opened unchanged v1.0; compare changed source rows with the reference.	Do not claim reference validation if the sample was modified.
Transaction marked CHECK	Inspect missing mandatory data, nonpositive hours, negative amounts and departure chronology.	Document source correction; never suppress the flag without explanation.
Unexpected OPEN / OVERDUE	Verify CONTROL B7, date type, due date and paid date.	Recalculate as-of classification after correction.
Alternative results appear zero	Check alternative ticket refs and timestamps; absence is not performance deterioration.	Do not infer change from missing comparison fields.
Unexpected cost/margin	Inspect quantity, quoted rate, invoice amount and the three direct-cost categories.	Do not classify contribution as net income.
Fuel estimates conflict	Confirm liters, equipment hours, period, asset type and independent-ledger status.	Do not combine unreconciled sample pools.

Release review sign-off (for a future controlled version)

Review check	Disposition
Authorized source data and documented provenance	[ ] Accepted [ ] Not yet satisfied
Versioned dictionary, formulas and sample controls reviewed	[ ] Accepted [ ] Not yet satisfied
Safety, confidentiality and regulatory limits assessed	[ ] Accepted [ ] Not yet satisfied
Review of failures, missingness, and confounding factors	[ ] Accepted [ ] Not yet satisfied
Independent reviewer identified and conclusions retained	[ ] Accepted [ ] Not yet satisfied

Reviewer name: _____ Review date: _____

Release disposition: [ ] Reference demonstration only [ ] Field pilot approved under separate protocol

Appendix A Source and evidence notices

Reference package

Workbook: PCSPF_v1_0_US_Operations_Toolkit.xlsx

SHA-256 of the workbook supplied with this guide:

cce3949a552712b0cd375b0f193c3150e7ad29d6ad7b787ce4eb7de877fd6704

Professional-plan basis: Ricardo Lopes Wiest, Port-Corridor SME Performance Framework (PCSPF), Professional Plan and Integrated Evidence Dossier, November 2026, especially Chapters 4, 15, 16, 22 and 23.

Definition of release status

“Prototype” means an implementation of selected calculations and document conventions, not a quality-certified or independently audited financial system. “Synthetic” means fabricated or mathematically generated data that do not describe any actual customer, employer, worker, U.S. company or realized operational improvement. “Field pilot” means an authorized, documented use with real records; no such pilot is represented here.

External context, not endorsement

U.S. Department of Transportation, National Freight Strategic Plan: https://www.transportation.gov/freight/NFSP

National Institute of Standards and Technology, Manufacturing Extension Partnership, Lean and Process Improvement: https://www.nist.gov/mep/lean-and-process-improvement

These references provide industry background only. PCSPF is not sponsored, certified, affiliated with or endorsed by USDOT, NIST, USCIS or any other U.S. agency.

Use in an immigration evidence record

The dated workbook and this guide may support a factual claim that a technical demonstration was prepared. Neither document establishes that the endeavor has achieved national importance, that an American company adopted PCSPF, that a reviewer endorsed it or that a petition has been approved. Any later assertion of impact should cite separately authenticated third-party and field evidence.

RELEASE STATEMENT

PCSPF v1.0 is supplied for reproducible internal demonstration and independent technical examination, with no implied commercial warranty, financial attestation, field effect or immigration determination.